Human innate Immunosenescence: causes and consequences for immunity in old age

The past decade has seen an explosion in research focusing on innate immunity. Through a wide range of mechanisms including phagocytosis, intracellular killing, and activation of pro-inflammatory or antiviral cytokine production via pattern recognition receptors, the cells of the innate immune system initiate and support adaptive immunity. The effects of aging on innate immune responses remain incompletely understood, particularly in humans. Here, we review advances in the study of human immunosenescence in the diverse cells of the innate immune system, including neutrophils, monocytes, macrophages, NK and NKT cells, and dendritic cells—with a focus on consequences for the response to infection or vaccination in old age.